

Human Anatomy and Physiology

Body systems, organs, and homeostasis

This material is designed as a concise student handout. It introduces core ideas, important vocabulary, and short review questions for classroom or self-study use.

Learning outcomes

- Explain the core concepts using accurate biology terminology.
- Connect biological structures to their functions.
- Apply the concepts to simple examples and review questions.
- Use the vocabulary section to strengthen scientific communication.

1. Organization of the body

The human body is organized into levels: cells, tissues, organs, organ systems, and the whole organism. Each level depends on the level below it.

A tissue is a group of similar cells working together. An organ contains different tissues that cooperate to perform a function.

Key points

- Cells form tissues.
- Tissues form organs.
- Organs work together in organ systems.

Mini activity

Choose one organ and identify at least two tissues that help it function.

2. Major body systems

The circulatory system transports oxygen, nutrients, hormones, and wastes. The respiratory system exchanges oxygen and carbon dioxide with the environment.

The digestive system breaks down food and absorbs nutrients. The nervous and endocrine systems help coordinate body responses.

Key points

- Circulatory system: transport.
- Respiratory system: gas exchange.
- Digestive system: nutrient processing.
- Nervous system: rapid control and communication.

Mini activity

Make a two-column chart matching body systems with their main functions.

3. Homeostasis

Homeostasis is the maintenance of a stable internal environment. The body constantly adjusts temperature, blood sugar, water balance, and other conditions.

Negative feedback is a common control mechanism. When a condition moves away from normal, the body responds to bring it back toward the normal range.

Key points

- Homeostasis keeps internal conditions stable.
- Negative feedback reverses changes from a set point.
- Sweating helps cool the body when temperature rises.

Mini activity

Explain how sweating and shivering help regulate body temperature.

Important vocabulary

Term	Meaning
Tissue	Group of similar cells working together.
Organ	Body structure made of multiple tissues.
Organ system	Group of organs that cooperate for major functions.
Homeostasis	Maintenance of stable internal conditions.
Negative feedback	Control process that reduces deviation from normal.

Quick review quiz

1. What are the levels of organization in the human body?
2. Which system transports oxygen to body cells?
3. How do the respiratory and circulatory systems work together?
4. What is homeostasis?
5. Give one example of negative feedback in the human body.

Teacher note: Students should answer the review questions before checking a textbook, notes, or class discussion.